

Conjecture 00000003891 is false: the ghost inverse problem is not well posed

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1 The conjecture

Definition: The ghost components w_n of Witt vectors are the weighted power sums $\sum x_i^{p^i}$. *Conjecture:* The recursion recovering Witt components from a ghost sequence always has a unique solution, and ghost sequences making component p -adic valuations unbounded form a measure-zero closed set in the compact topology.

The claim is a conjunction: (I) for every ghost sequence the recursion has exactly one solution, and (II) the set S of ghost sequences whose recovered components have unbounded p -adic valuations is closed *and* has measure zero. We refute it for every prime p under each of the readings below. All statements are proved in Lean 4 / Mathlib (file `Conjecture3891/Basic.lean`, namespace `C3891`).

2 Definitions and conventions

Definition 1 (Ghost components). For a commutative ring R and a p -typical Witt vector $x = (x_0, x_1, \dots)$ over R (Mathlib’s `WittVector p R`, components `x.coeff i`), the n -th ghost component is

$$w_n(x) = \sum_{i=0}^n p^i x_i^{p^{n-i}}.$$

This is Mathlib’s `WittVector.ghostComponent n x`; the displayed formula is `ghost_formula`, proved from `WittVector.ghostComponent_apply` and `aeval_wittPolynomial`. The ghost map is `WittVector.ghostMap`. The recursion of the conjecture is $p^n x_n = w_n - \sum_{i < n} p^i x_i^{p^{n-i}}$.

The printed formula $\sum x_i^{p^i}$ has no weights, although the text calls the ghost components “weighted power sums”; we take the named object (ghost components of Witt vectors) as primary and treat the literal formula separately in Section 6.

Ambient space. Ghost sequences range over $\mathbb{Z}_p^{\mathbb{N}} = \mathbb{N} \rightarrow \mathbb{Z}_p$ with the product topology, which is compact (the “compact topology”). Mathlib has no `MeasurableSpace` instance on $\mathbb{Z}_p$; we prove the measure statement for *every* σ -algebra on $\mathbb{Z}_p^{\mathbb{N}}$ and every measure that is positive on nonempty open sets. In particular it applies to every additive Haar measure for the Borel σ -algebra, e.g. the Haar probability measure `haarP p = Measure.addHaarMeasure ⊤ (haar_ne_zero)`.

Valuations. $v = \text{Padic.valuation}$ on $\mathbb{Q}_p$, with $\|y\| = p^{-v(y)}$ for $y \neq 0$. Mathlib sets `valuation 0 = 0`; our sets never use this value: they either exclude zero components or count them explicitly as $+\infty$. With $x = x(w)$ the recovered components:

- `UnbddBelow p`: for every $M \in \mathbb{Z}$ some n has $x_n \neq 0$, $v(x_n) < M$;

- **Unbdd p**: for every M some n has $x_n \neq 0$, $M < |v(x_n)|$;
- **UnbddAbove p**: for every M some n has $x_n \neq 0$, $M < v(x_n)$;
- **UnbddAboveTop p**: for every M some n has $x_n = 0$ or $M < v(x_n)$ (zero counts as $+\infty$).

Here, for $w \in \mathbb{Z}_p^{\mathbb{N}}$, $x(w) = \text{comps } p \ w$ is the coefficient sequence of the unique Witt vector over $\mathbb{Q}_p$ with ghost sequence w : $\text{cvec } p \ w = (\text{WittVector.ghostEquiv } p \ \mathbb{Q}_p[p]).\text{symm } w$ (ghost map bijective since p is invertible in $\mathbb{Q}_p$); uniqueness is **cvec_unique**. For w in the ghost image of $W(\mathbb{Z}_p)$ these are its integral components (**comps_ghost**).

3 Reading A: components in $\mathbb{Z}_p$

Lemma 2 (**e1_not_ghost**). *The ghost sequence $e = (0, 1, 0, 0, \dots)$ is not $\text{ghostMap}(x)$ for any $x \in W(\mathbb{Z}_p)$.*

Proof. $w_0 = x_0 = 0$, so $w_1 = x_0^p + px_1 = px_1 = 1$; but $\|px_1\| \leq p^{-1} < 1 = \|1\|$. $\square$

Hence clause (I) fails over $\mathbb{Z}_p$: $\neg \forall w, \exists! x : \text{WittVector } p \ \mathbb{Z}_p[p], \text{ghostMap } x = w$.

4 Reading B: components in $\mathbb{Q}_p$

Over $\mathbb{Q}_p$ clause (I) holds (part of **main_theorem**). Let $a_n = 1 + p + \dots + p^{n-1} = (p^n - 1)/(p - 1)$ (**aexp**: $a_0 = 0$, $a_{n+1} = pa_n + 1$).

Lemma 3 (**key**, **aexp_ge**). *For all i, m : $a_i p^{m+2} + m + 1 < a_{i+m+1} p$; and $n \leq a_n$.*

Proof. Induction on m : for $m = 0$, $(pa_i + 1)p = p^2 a_i + p > p^2 a_i + 1$. Step: multiply the hypothesis by p and use $m + 2 \leq (m + 1)p$ and $a_{i+m+2}p = p^2 a_{i+m+1} + p$. $\square$

Theorem 4 (**norm_comps**, **comps_ne_zero_val**). *If $w \in \mathbb{Z}_p^{\mathbb{N}}$ and $\|w_1 - w_0^p\| = 1$, then for all $n \geq 1$: $\|x_n\| = p^{a_n}$, so $x_n \neq 0$ and $v(x_n) = -(p^n - 1)/(p - 1)$.*

Proof. Strong induction. $n = 1$: $px_1 = w_1 - w_0^p$ gives $\|x_1\| = p$. For $n + 2$: write $p^{n+2}x_{n+2} = (w_{n+2} - S) - D$ with $D = p^{n+1}x_{n+1}^p$, $\|D\| = p^{a_{n+1}p - (n+1)}$, and $S = \sum_{i \leq n} p^i x_i^{p^{n+2-i}}$. By induction $\|x_i\| \leq p^{a_i}$ for $i \leq n + 1$ (for $i = 0$, $x_0 = w_0$), so each term of S has norm $\leq p^{a_i p^{n+2-i} - i} < \|D\|$ by **key** with $m = n - i$; by the ultrametric inequality $\|S\| < \|D\|$, and $\|w_{n+2}\| \leq 1 < \|D\|$ since $a_{n+1}p > n + 1$. Hence $\|p^{n+2}x_{n+2}\| = \|D\|$ and $\|x_{n+2}\| = p^{a_{n+1}p+1} = p^{a_{n+2}}$. $\square$

Let $U = \{w : \|w_1 - w_0^p\| = 1\}$. It is open (preimage of the sphere of radius 1, open in an ultrametric space: **isOpen_U**) and nonempty ($e \in U$). The theorem gives $U \subseteq \text{UnbddBelow } p \ (\text{U_subset})$, and $\text{UnbddBelow } p \subseteq \text{Unbdd } p$.

Theorem 5 (**measure_ne_zero**, **haar_ne_zero**). *For every σ -algebra on $\mathbb{Z}_p^{\mathbb{N}}$, every measure μ positive on nonempty open sets, and every $T \supseteq \text{UnbddBelow } p$: $\mu(T) \neq 0$. In particular for every additive Haar measure.*

So the unbounded-valuation set (unbounded below, or unbounded in absolute value) is not of measure zero, and clause (II) fails.

5 Reading C: “unbounded” means unbounded above

Let $x_n^{(k)} = 1$ for $n < k$ and $x_n^{(k)} = p^n$ for $n \geq k$ ($\mathbf{xk}$), $w^{(k)} = \text{ghost}(x^{(k)})$, $w^* = \text{ghost}(1, 1, 1, \dots)$. Since w_n depends only on $x_0, \dots, x_n$, $w_n^{(k)} = w_n^*$ for $k > n$, so $w^{(k)} \rightarrow w^*$ in the product topology ($\mathbf{tendsto_wk}$). By uniqueness the recovered components of $w^{(k)}$ are $x^{(k)}$, with $v(p^n) = n$, so $w^{(k)} \in \mathbf{UnbddAbove\ p\ (wk_mem)}$; those of w^* are all 1, valuation 0, so $w^* \notin \mathbf{UnbddAboveTop\ p\ (wstar_not_mem)}$.

Theorem 6 ($\mathbf{not_closed, not_closed_in}$). *Every T with $\mathbf{UnbddAbove\ p} \subseteq T \subseteq \mathbf{UnbddAboveTop\ p}$ is not closed in $\mathbb{Z}_p^{\mathbb{N}}$, and $T \cap A$ is not relatively closed in any $A \supseteq \text{ghost}(W(\mathbb{Z}_p))$, in particular in the ghost image itself (where clause (I) holds with integral components, $\mathbf{ghost_injective_Zp}$).*

6 The literal printed formula

If $w_n = \sum_{i \leq n} x_i^{p^i} (\mathbf{litGhost})$, the ghost sequence $(0, p, 0, \dots)$ forces $x_0 = 0$ and $x_1^p = p$, i.e. $p v(x_1) = 1$, impossible ($\mathbf{lit_no_solution}$); so clause (I) fails over $\mathbb{Q}_p$. (Since $\mathbb{Z}_p \subset \mathbb{Q}_p$, there is then no integral solution either; this last remark is not separately stated in Lean.)

7 Lean summary and axiom audit

$\mathbf{C3891.main_theorem}$ (for every prime p , $[\mathbf{Fact\ p.Prime}]$) is the conjunction of: (A) $\neg \forall w : \mathbb{N} \rightarrow \mathbb{Z}_p, \exists! x : \mathbf{WittVector\ p\ \mathbb{Z}_p}, \mathbf{ghostMap\ x} = w$; (B) unique solvability over $\mathbb{Q}_p$, and $\mu(T) \neq 0$ for every open-positive μ on any σ -algebra and every $T \supseteq \mathbf{UnbddBelow\ p}$, together with $\mathbf{UnbddBelow\ p} \subseteq \mathbf{Unbdd\ p}$; (C) for every T between $\mathbf{UnbddAbove\ p}$ and $\mathbf{UnbddAboveTop\ p}$, $\neg \mathbf{IsClosed\ T}$ and $\neg \mathbf{IsClosed\ (Subtype.val^{-1}\ T)}$ in $\mathbf{ghostImage\ p}$; (L) the literal formula is not always solvable over $\mathbb{Q}_p$. The file is 411 lines and imports only `Mathlib`. `lake build` succeeds and

```
'C3891.main_theorem' depends on axioms: [propext, Classical.choice, Quot.sound]
'C3891.haar_ne_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
```

No sorry, no `native_decide`, no added axioms.

Readings not covered. Ghost sequences ranging over the non-compact space $\mathbb{Q}_p^{\mathbb{N}}$ are not treated (the text specifies a compact topology). Under the integral reading restricted to the ghost image with “unbounded” meaning unbounded *below*, the set is empty (integral components have $v \geq 0$); this degenerate reading is not refuted.